

Q1 Let $u = (u_1, u_2)$ and $v = (v_1, v_2)$ belong to $\mathbb{R}^2$.

verify that $\langle u, v \rangle = u_1 v_1 - 2u_1 v_2 - 2u_2 v_1 + 5u_2 v_2$ is an I.P. on $\mathbb{R}^2$.

$$(i) \langle u, u \rangle = u_1 u_1 - 2u_1 u_2 - 2u_2 u_1 + 5u_2 u_2$$

$$= u_1^2 - 4u_1 u_2 + 5u_2^2$$

$$= u_1^2 - 4u_1 u_2 + (2u_1^2) + (2u_2^2) + 5u_2^2$$

$$= (u_1 - 2u_2)^2 + u_2^2 \geq 0$$

completing square.

$$\langle u, u \rangle = 0$$

$$\Leftrightarrow (u_1 - 2u_2)^2 + u_2^2 = 0$$

$$\Leftrightarrow (u_1 - 2u_2)^2 = 0 \text{ and } u_2^2 = 0$$

$$\Leftrightarrow u_1 - 2u_2 = 0 \text{ and } u_2 = 0$$

$$\Leftrightarrow u_1 - 2(0) = 0 \therefore u_1 = 0$$

$$\Leftrightarrow u_1 = 0$$

$$\Leftrightarrow u = 0 \therefore u = (u_1, u_2)$$

$$(ii) \langle u, v \rangle = u_1 v_1 - 2u_1 v_2 - 2u_2 v_1 + 5u_2 v_2$$

$$= v_1 u_1 - 2v_2 u_1 - 2v_1 u_2 + 5v_2 u_2$$

$$= \langle v, u \rangle$$

$$au + bv = (au + bv)_1, (au + bv)_2 \\ w = (w_1, w_2)$$

$$(iii) \langle au + bv, w \rangle = (au + bv)_1 w_1 - 2(au + bv)_1 w_2 - 2(au + bv)_2 w_1 + 5(au + bv)_2 w_2$$

$$= au_1 w_1 + bv_1 w_1 - 2au_1 w_2 - 2bv_1 w_2 - 2au_2 w_1 - 2bv_2 w_1 + 5au_2 w_2 + 5bv_2 w_2$$

$$= a(u_1 w_1 - 2u_1 w_2 - 2u_2 w_1 + 5u_2 w_2) + b(v_1 w_1 - 2v_1 w_2 - 2v_2 w_1 + 5v_2 w_2)$$

$$= a \langle u, w \rangle + b \langle v, w \rangle$$

All the three conditions for I.P. are satisfied

Hence $\langle u, v \rangle$ as defined above is an I.P. on $\mathbb{R}^2$.

Q_{1(iii)} For what value of K $\langle u, v \rangle = u_1 v_1 - 3u_1 v_2 - 3u_2 v_1 + K u_2 v_2$ is an I.P. on $\mathbb{R}^2$.

Sol

$$\begin{aligned}\langle u, u \rangle &= u_1 u_1 - 3u_1 u_2 - 3u_2 u_1 + K u_2 u_2 \\ &= u_1^2 - 6u_1 u_2 + K u_2^2\end{aligned}$$

$$\begin{aligned}u &= (u_1, u_2) \\ v &= (v_1, v_2)\end{aligned}$$

For an I.P. condition (i) i.e. $\langle u, u \rangle$ must be +ve
for this $\langle u, u \rangle$ must be in perfect square form

$$\begin{aligned}\therefore \langle u, u \rangle &= u_1^2 - 6u_1 u_2 + 9u_2^2 \text{ is perfect square form for } K \geq 9 \\ &= (u_1 - 3u_2)^2 \text{ for } K = 9 \\ &= (u_1 - 3u_2)^2 + u_2^2 \text{ for } K = 10\end{aligned}$$

Cond (ii) & (iii) are also satisfied for $K \geq 9$.

Q₂ Find the norm of $(2, 3) \in \mathbb{R}^2$ w.r.t. Euclidean inner product on $\mathbb{R}^2$

(i) $\langle u, v \rangle = u_1 v_1 + u_2 v_2$ $u = (u_1, u_2)$ $v = (v_1, v_2)$

$$\|u\| = \sqrt{\langle u, u \rangle} = \sqrt{u_1^2 + u_2^2}$$

$$\|(2, 3)\| = \sqrt{2^2 + 3^2} = \sqrt{13} \text{ Ans}$$

(ii) I.P. defined $\langle u, v \rangle = u_1 v_1 - 2u_1 v_2 - 2u_2 v_1 + 5u_2 v_2$

$$\begin{aligned}\|u\| &= \sqrt{\langle u, u \rangle} = \sqrt{u_1 u_1 - 2u_1 u_2 - 2u_2 u_1 + 5u_2 u_2} \\ &= \sqrt{u_1^2 - 4u_1 u_2 + 5u_2^2}\end{aligned}$$

$$\|(2, 3)\| = \sqrt{2^2 - 4 \cdot 2 \cdot 3 + 5 \cdot 3^2}$$

$$= \sqrt{4 - 24 + 45}$$

$$= \sqrt{25} = 5 \text{ Ans}$$

Q₃ Find the norm of $(\frac{1}{2}, -\frac{1}{4}, \frac{1}{3}, \frac{1}{6}) \in \mathbb{R}^4$ w.r.t. Euclidean I.P. on $\mathbb{R}^4$

$\langle u, v \rangle = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4$ Euclidean I.P. on $\mathbb{R}^4$

$$\begin{aligned}\|u\| &= \sqrt{\langle u, u \rangle} \\ &= \sqrt{u_1^2 + u_2^2 + u_3^2 + u_4^2}\end{aligned}$$

$$\|(\frac{1}{2}, -\frac{1}{4}, \frac{1}{3}, \frac{1}{6})\| = \sqrt{(\frac{1}{2})^2 + (-\frac{1}{4})^2 + (\frac{1}{3})^2 + (\frac{1}{6})^2}$$

$$= \sqrt{\frac{1}{4} + \frac{1}{16} + \frac{1}{9} + \frac{1}{36}}$$

$$= \sqrt{\frac{36+9+16+4}{144}} = \sqrt{\frac{65}{144}} = \frac{\sqrt{65}}{12} \text{ Ans}$$

Q4 Let V denote the vector space of 2×2 matrices over $\mathbb{R}$. If $A, B \in V$ and $\text{Tr}(A)$ (called the Trace of A) denotes the sum of diagonal elements of A , show that $\langle A, B \rangle = \text{Tr}(B^t A)$ is an I.P. on V . Also find

$$\text{norm of } A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$$

Sol Let $A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$ $B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$

$$B^t A = \begin{bmatrix} b_1 & b_3 \\ b_2 & b_4 \end{bmatrix} \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$$

$$B^t A = \begin{bmatrix} b_1 a_1 + b_3 a_3 & b_1 a_2 + b_3 a_4 \\ b_2 a_1 + b_4 a_3 & b_2 a_2 + b_4 a_4 \end{bmatrix}$$

$$\langle A, B \rangle = \text{Tr}(B^t A)$$

$$= b_1 a_1 + b_2 a_2 + b_3 a_3 + b_4 a_4 \quad \text{(sum of diagonal element)} \quad \textcircled{1}$$

$$\begin{aligned} \text{(i)} \quad \langle A, A \rangle &= a_1^2 + a_2^2 + a_3^2 + a_4^2 \\ &= a_1^2 + a_2^2 + a_3^2 + a_4^2 \geq 0 \end{aligned}$$

$$\text{If } \langle A, A \rangle = 0 \Rightarrow a_1^2 + a_2^2 + a_3^2 + a_4^2 = 0$$

$$\Rightarrow a_1^2 = 0, a_2^2 = 0, a_3^2 = 0, a_4^2 = 0$$

$$\Rightarrow a_1 = 0, a_2 = 0, a_3 = 0, a_4 = 0$$

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

$$\begin{aligned} \text{(ii)} \quad \langle A, B \rangle &= b_1 a_1 + b_2 a_2 + b_3 a_3 + b_4 a_4 \\ &= a_1 b_1 + a_2 b_2 + a_3 b_3 + a_4 b_4 \\ &= \langle B, A \rangle \end{aligned}$$

L, M are const.

$$\text{(iii)} \quad \langle LA + MB, C \rangle = \text{Tr}(C^t (LA + MB))$$

$$C = \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix}$$

$$= c_1(La_1 + Mb_1) + c_2(La_2 + Mb_2) + c_3(La_3 + Mb_3) + c_4(La_4 + Mb_4) \quad \text{using } \textcircled{1}$$

$$= L(c_1 a_1 + c_2 a_2 + c_3 a_3 + c_4 a_4) + M(c_1 b_1 + c_2 b_2 + c_3 b_3 + c_4 b_4)$$

$$= L \langle A, C \rangle + M \langle B, C \rangle \quad \text{Hence } \langle A, B \rangle \text{ is I.P. on } V.$$

$$\|A\| = \sqrt{\langle A, A \rangle} = \sqrt{a_1^2 + a_2^2 + a_3^2 + a_4^2}$$

$$\left\| \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} \right\| = \sqrt{1^2 + 2^2 + 3^2 + (-4)^2} = \sqrt{1+4+9+16} = \sqrt{30}$$

Q5 Let V be the vector space $P(n)$ of polynomials over $\mathbb{R}$. Show that $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$ defines an inner product on V .

Sol. i) $\langle f, f \rangle = \int_0^1 f(t)f(t)dt$
 $= \int_0^1 f^2(t)dt \geq 0$ as definite integral represents area of plane region.

if $\langle f, f \rangle = 0$

$\Leftrightarrow \int_0^1 f^2(t)dt = 0$

$\Leftrightarrow f^2(t) = 0$

$\Leftrightarrow f(t) = 0$

ii) $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$

$= \int_0^1 g(t)f(t)dt$

$= \langle g, f \rangle$

iii) $\langle af+bg, h \rangle = \int_0^1 (af+bg)h(t)dt$

$= a \int_0^1 f(t)h(t)dt + b \int_0^1 g(t)h(t)dt$

$= a \langle f, h \rangle + b \langle g, h \rangle$

Hence $\langle f, g \rangle$ is I.P in V .

⑥ Let $U_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ $U_2 = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$ $U_3 = \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix}$

Using Example 2, find i.e. $\langle U_i, U_j \rangle = \det(U_i^t U_j)$

- the inner product of each pair of above column vectors
- the norm of each vector
- a vector orthogonal to U_1 & U_2
- a vector orthogonal to U_1 & U_3

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Sol (i) $\langle U_1, U_2 \rangle = \det(U_1^t U_2) = \det \left(\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \right) = \det \begin{bmatrix} 1(2) + 2(1) + 1(2) \end{bmatrix} = \det[6] = 6$

$$\langle U_1, U_3 \rangle = \det(U_1^t U_3) = \det \left(\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix} \right) = \det \begin{bmatrix} 1(2) + 2(1) + 1(-4) \end{bmatrix} = 0$$

$$\langle U_2, U_3 \rangle = \det(U_2^t U_3) = \det \left(\begin{bmatrix} 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix} \right) = \det \begin{bmatrix} 2(2) + 1(1) + 2(-4) \end{bmatrix} = -3$$

(ii) $\|U_1\| = \sqrt{\langle U_1, U_1 \rangle} = \sqrt{\det(U_1^t U_1)} = \sqrt{\det \left(\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right)} = \sqrt{\det \begin{bmatrix} 1^2 + 2^2 + 1^2 \end{bmatrix}} = \sqrt{\det[6]} = \sqrt{6}$

$$\|U_2\| = \sqrt{\langle U_2, U_2 \rangle} = \sqrt{\det(U_2^t U_2)} = \sqrt{\det \left(\begin{bmatrix} 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \right)} = \sqrt{\det \begin{bmatrix} 2^2 + 1^2 + 2^2 \end{bmatrix}} = \sqrt{\det[9]} = 3$$

$$\|U_3\| = \sqrt{\langle U_3, U_3 \rangle} = \sqrt{\det(U_3^t U_3)} = \sqrt{\det \left(\begin{bmatrix} 2 & 1 & -4 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix} \right)} = \sqrt{\det \begin{bmatrix} 2^2 + 1^2 + 4^2 \end{bmatrix}} = \sqrt{\det[21]} = \sqrt{21}$$

(iii) Suppose $U = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ be a vector orthogonal to U_1 & U_2

$$\therefore \langle U, U_1 \rangle = 0 \Rightarrow \det(U^t U_1) = 0 \Rightarrow \det \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = 0$$

$$\Rightarrow x + 2y + z = 0 \quad \text{--- (I)}$$

$$\langle U, U_2 \rangle = 0 \Rightarrow \det(U^t U_2) = 0 \Rightarrow \det \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = 0$$

$$\Rightarrow 2x + y + 2z = 0 \quad \text{--- (II)}$$

from (I) & (II) $x + 2y + z = 0$
 $2x + y + 2z = 0$

$$\frac{x}{4-1} = \frac{-y}{2-2} = \frac{z}{1-4} = K$$

$$x = 3K$$

$$y = 0$$

$$z = -3K$$

$$U = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 3K \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \text{ is orthogonal to } U_1 \text{ & } U_2$$

(iv) Let $U = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ be a vector orthogonal to U_1 & U_3

$$\therefore \langle U, U_1 \rangle = x + 2y + z = 0 \quad \text{--- (I)}$$

$$\langle U, U_3 \rangle = 2x + y - 4z = 0 \quad \text{--- (II)}$$

$$\frac{x}{-8-1} = \frac{-y}{+4-2} = \frac{z}{1-4} = K$$

$$x = -9K$$

$$y = +6K$$

$$z = -3K$$

$$U = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -3K \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$$

is orthogonal to U_1 & U_3

2nd Method

$$\begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ 2 & 1 & 2 \end{vmatrix}$$

$$= 3i - 0j + -3k$$

$$= 3(i - 0j - k)$$

$$\therefore \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \text{ is required vector}$$

2nd Method

$$\begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ 2 & 1 & -4 \end{vmatrix}$$

$$= -9i - j(-6) + 11(1-2)$$

$$= -3(3i + 2j + k)$$

$$\therefore \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} \text{ is orthogonal to } U_1 \text{ & } U_3$$

(i) Show that $(1,1), (0,1)$ is a basis of $\mathbb{R}^2$. Using the Gram-Schmidt process, find an orthonormal basis of $\mathbb{R}^2$.

Let $(x,y) \in \mathbb{R}^2$ $S = \{(1,1), (0,1)\}$

Consider $(x,y) = a(1,1) + b(0,1)$

$(x,y) = (a, a+b)$

Linear Combination
of $a, b \in \mathbb{R}$ given vectors

$\Rightarrow \boxed{a=x}$

$\neq a+b=y \Rightarrow x+b=y \Rightarrow \boxed{b=y-x}$

$\therefore$ the linear combination exists, as $(1,1) \neq (0,1)$ can be written as a linear combination of a & b . So the set 'S' is spanning set for $\mathbb{R}^2$.

Now for Linear Independence.

$a(1,1) + b(0,1) = 0$

$\boxed{a=0}$

$a+b=0 \Rightarrow 0+b=0 \Rightarrow \boxed{b=0}$

Hence given vectors are Linearly Independent.

S is spanning set & Linearly Independent. So S is basis for $\mathbb{R}^2$.

Orthonormal Basis of $\mathbb{R}^2$ by Gram-Schmidt Process.

Let $v_1 = (1,1)$ $v_2 = (0,1)$

Now $u_1 = \frac{v_1}{\|v_1\|} = \frac{(1,1)}{\sqrt{1^2+1^2}} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

$u_2 = \frac{w_2}{\|w_2\|}$ where $w_2 = v_2 - \langle v_2, u_1 \rangle u_1$
 $= (0,1) - \langle (0,1), \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \rangle \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
 $= (0,1) - \left(0 \cdot \frac{1}{\sqrt{2}} + 1 \cdot \frac{1}{\sqrt{2}}\right) \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
 $= (0,1) - \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
 $= (0,1) - \left(\frac{1}{2}, \frac{1}{2}\right)$
 $= \left(0 - \frac{1}{2}, 1 - \frac{1}{2}\right)$

$w_2 = \left(-\frac{1}{2}, \frac{1}{2}\right)$

$\|w_2\| = \sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{1}{4}} = \frac{1}{\sqrt{2}}$

$\therefore u_2 = \frac{\left(-\frac{1}{2}, \frac{1}{2}\right)}{\frac{1}{\sqrt{2}}}$

$u_2 = \sqrt{2} \left(-\frac{1}{2}, \frac{1}{2}\right) = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

Thus the orthonormal basis is $\{u_1, u_2\} = \left\{\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)\right\}$

$$\|\omega_3\| = \sqrt{1^2 + 0^2 + 0^2} = 1$$

$$u_3 = \frac{\omega_3}{\|\omega_3\|} = \frac{(1, 0, 0)}{1} = (1, 0, 0)$$

Therefore orthonormal basis is $u_1, u_2, u_3 = \{(0, 0, 1), (0, 1, 0), (1, 0, 0)\}$

x-----x

(ii) Taking $v_1 = (1, 0, 1)$, $v_2 = (0, 1, 1)$, $v_3 = (0, 0, 1)$

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{(1, 0, 1)}{\sqrt{1^2 + 0^2 + 1^2}} = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$$

$$\begin{aligned} u_2 &= \frac{\omega_2}{\|\omega_2\|} \text{ where } \omega_2 = v_2 - \langle v_2, u_1 \rangle u_1 \\ &= (0, 1, 1) - \langle (0, 1, 1), \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) \rangle \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) \\ &= (0, 1, 1) - \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) \\ &= (0, 1, 1) - \left(\frac{1}{2}, 0, \frac{1}{2}\right) \end{aligned}$$

$$\omega_2 = \left(-\frac{1}{2}, 1, \frac{1}{2}\right)$$

$$\|\omega_2\| = \sqrt{\frac{1}{4} + 1 + \frac{1}{4}} = \sqrt{\frac{6}{4}} = \sqrt{\frac{3}{2}}$$

$$\begin{aligned} \therefore u_2 &= \frac{\omega_2}{\|\omega_2\|} = \frac{\left(-\frac{1}{2}, 1, \frac{1}{2}\right)}{\sqrt{\frac{3}{2}}} = \frac{\sqrt{2}}{\sqrt{3}} \left(-\frac{1}{2}, 1, \frac{1}{2}\right) = \left(-\frac{1}{\sqrt{6}}, \frac{\sqrt{2}}{\sqrt{3}}, \frac{1}{\sqrt{6}}\right) \\ &= \left(-\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right) \end{aligned}$$

$$\begin{aligned} u_3 &= \frac{\omega_3}{\|\omega_3\|} \text{ where } \omega_3 = v_3 - \langle v_3, u_1 \rangle u_1 - \langle v_3, u_2 \rangle u_2 \\ &= (0, 0, 1) - \langle (0, 0, 1), \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) \rangle \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) \\ &\quad - \langle (0, 0, 1), \left(-\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right) \rangle \left(-\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right) \\ &= (0, 0, 1) - \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) - \frac{1}{\sqrt{6}} \left(-\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right) \\ &= (0, 0, 1) - \left(\frac{1}{2}, 0, \frac{1}{2}\right) - \left(-\frac{1}{6}, \frac{2}{6}, \frac{1}{6}\right) \\ &= \left(0 - \frac{1}{2}, 0, \frac{1}{2}\right) - \left(-\frac{1}{6}, \frac{2}{6}, \frac{1}{6}\right) \end{aligned}$$

$$\omega_3 = \left(-\frac{1}{2} + \frac{1}{6}, -\frac{2}{6}, \frac{1}{2} - \frac{1}{6}\right) = \left(-\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}\right)$$

$$\|\omega_3\| = \sqrt{\frac{1}{9} + \frac{1}{9} + \frac{1}{9}} = \sqrt{\frac{3}{9}} = \sqrt{\frac{1}{3}}$$

$$u_3 = \frac{\omega_3}{\|\omega_3\|} = \frac{\left(-\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}\right)}{\frac{1}{\sqrt{3}}} = \sqrt{3} \left(-\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}\right) = \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

Show that $(1,0,1), (0,1,1), (0,0,1)$ is a basis of R^3 . Using the Gram Schmidt process, find an orthonormal basis of R^3 by taking $u_1 = (0,0,1)$

$$\text{Let } S = \{(1,0,1), (0,1,1), (0,0,1)\} \subset R^3$$

$$\text{Let } (x,y,z) \in R^3$$

$$\begin{aligned} \text{Consider } (x,y,z) &= a(1,0,1) + b(0,1,1) + c(0,0,1) \\ &= (a, b, a+b+c) \end{aligned}$$

$$\boxed{a=x}$$

$$\boxed{b=y}$$

$$a+b+c=z \Rightarrow \boxed{c=z-x-y} \quad \text{So linear combination of vectors of } S \text{ exists.} \\ \text{So } S \text{ is spanning set for } R^3.$$

$$\text{Now } a(1,0,1) + b(0,1,1) + c(0,0,1) = 0$$

To check Linear Independence.
2nd Method

$$a + 0 + 0 = 0 \Rightarrow \boxed{a=0}$$

$$0 + b + 0 = 0 \Rightarrow \boxed{b=0}$$

$$a + b + c = 0 \Rightarrow \boxed{c=0}$$

Hence S is Linear Independent.

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Rank of $A = 3 = \text{No. of vectors in } S.$
 $\therefore S$ is independent.

So S is a basis of R^3 as S is spanning & Linear Independent.

Gram Schmidt Orthonormalization Process.

$$\text{Taking } v_1 = (0,0,1) \quad v_2 = (0,1,1) \quad v_3 = (1,0,1)$$

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{(0,0,1)}{\sqrt{0^2+0^2+1^2}} = (0,0,1)$$

$$\begin{aligned} u_2 &= \frac{\omega_2}{\|\omega_2\|} \quad \text{where } \omega_2 = v_2 - \langle v_2, u_1 \rangle u_1 \\ &= (0,1,1) - \langle (0,1,1), (0,0,1) \rangle (0,0,1) \\ &= (0,1,1) - (0+0+1)(0,0,1) \\ &= (0,1,1) - (0,0,1) \\ \omega_2 &= (0,1,0) \end{aligned}$$

$$\|\omega_2\| = \sqrt{0^2+1^2+0^2} = 1$$

$$\therefore u_2 = \frac{(0,1,0)}{1} = (0,1,0)$$

$$\begin{aligned} u_3 &= \frac{\omega_3}{\|\omega_3\|} \quad \text{where } \omega_3 = v_3 - \langle v_3, u_1 \rangle u_1 - \langle v_3, u_2 \rangle u_2 \\ &= (1,0,1) - \langle (1,0,1), (0,0,1) \rangle (0,0,1) - \langle (1,0,1), (0,1,0) \rangle (0,1,0) \\ &= (1,0,1) - 1(0,0,1) - 0(0,1,0) \\ &= (1,0,1) - (0,0,1) \\ \omega_3 &= (1,0,0) \end{aligned}$$

⑨ Show that $\{(1, -1, 0), (2, -1, -2), (1, -1, -2)\}$ is a basis of $\mathbb{R}^3$

Find an orthonormal basis of $\mathbb{R}^3$ using Gram-Schmidt Process.

Sol $S = \{(1, -1, 0), (2, -1, -2), (1, -1, -2)\}$

Let $(x, y, z) \in \mathbb{R}^3$

Consider $(x, y, z) = a(1, -1, 0) + b(2, -1, -2) + c(1, -1, -2) \quad \text{--- (i)}$

$x = a + 2b + c \quad \text{--- (ii) from (i)}$

$y = -a - b - c \quad \text{--- (iii)}$

$z = -2b - 2c \quad \text{--- (iv)}$

from (ii) + (iii) $x + y = b$ Put in (iv)

$z = -2x - 2y - 2c \Rightarrow 2c = -2x - 2y - z$

$c = -x - y - \frac{z}{2}$ Put in

from (iii) $y = -a - (x + y) - (-x - y - \frac{z}{2})$

$a = -y - x - y + x + y + \frac{z}{2} \Rightarrow a = \frac{z}{2} - y$

Linear combination of a, b, c & vectors of S exist. So S is spanning set.

Now

$a(1, -1, 0) + b(2, -1, -2) + c(1, -1, -2) = 0$

$a + 2b + c = 0 \quad \text{--- (v)}$

$-a - b - c = 0 \quad \text{--- (vi)}$

$-2b + (-2c) = 0 \Rightarrow b = \frac{-2c}{-2} = -c \quad \text{--- (vii)}$

$-a - b - c = 0 \Rightarrow -a - (-c) - c = 0$

$\Rightarrow -a + c - c = 0$

$\Rightarrow a = 0$ Put in (v)

$a + 2b + c = 0 \Rightarrow 0 + 2(-c) + c = 0$

$\Rightarrow c = 0$

by Put $c = 0$ in (vii) $b = 0$

Hence S is linearly independent.

" S is spanning & linearly independent
So S is basis of $\mathbb{R}^3$.

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2nd Method

$A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & -1 & -2 \\ 1 & -1 & -2 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & -2 \end{bmatrix}$ Echelon form

Rank of $A = 3 = N$ of vectors

So S is linearly independent

Gram Schmidt Process.

$$S = \{(1, -1, 0), (2, -1, -2), (1, -1, -2)\}$$

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{(1, -1, 0)}{\sqrt{1^2 + 1^2 + 0}} = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right)$$

$$\begin{aligned} \text{Let } u_2 = \frac{\omega_2}{\|\omega_2\|} \quad \text{where } \omega_2 &= v_2 - \langle v_2, u_1 \rangle u_1 \\ &= (2, -1, -2) - \langle (2, -1, -2), \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right) \rangle \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right) \\ &= (2, -1, -2) - \left(\frac{2}{\sqrt{2}} + \frac{1}{\sqrt{2}} + 0\right) \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right) \\ &= (2, -1, -2) - \frac{3}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right) \\ &= (2, -1, -2) - \left(\frac{3}{2}, -\frac{3}{2}, 0\right) \end{aligned}$$

$$\omega_2 = \left(\frac{1}{2}, \frac{1}{2}, -2\right)$$

$$\begin{aligned} \|\omega_2\| &= \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + (-2)^2} = \sqrt{\frac{1}{4} + \frac{1}{4} + 4} = \sqrt{\frac{18}{4}} \\ &= \sqrt{\frac{9}{2}} = \frac{3}{\sqrt{2}} \end{aligned}$$

$$\therefore u_2 = \frac{\omega_2}{\|\omega_2\|} = \frac{\left(\frac{1}{2}, \frac{1}{2}, -2\right)}{\frac{3}{\sqrt{2}}} = \frac{\sqrt{2}}{3} \left(\frac{1}{2}, \frac{1}{2}, -2\right) = \left(\frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, -\frac{2\sqrt{2}}{3}\right)$$

$$\begin{aligned} u_3 = \frac{\omega_3}{\|\omega_3\|} \quad \text{where } \omega_3 &= v_3 - \langle v_3, u_1 \rangle u_1 - \langle v_3, u_2 \rangle u_2 \\ &= (1, -1, -2) - \langle (1, -1, -2), \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right) \rangle \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right) \\ &\quad - \langle (1, -1, -2), \left(\frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, -\frac{2\sqrt{2}}{3}\right) \rangle \left(\frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, -\frac{2\sqrt{2}}{3}\right) \\ &= (1, -1, -2) - \left(\frac{1}{2} + \frac{1}{2} + 0\right) \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right) - \left(\frac{1}{3\sqrt{2}} - \frac{1}{3\sqrt{2}} + \frac{4\sqrt{2}}{3}\right) \left(\frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, -\frac{2\sqrt{2}}{3}\right) \\ &= (1, -1, -2) - \left(\frac{2}{\sqrt{2}}\right) \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right) - \left(\frac{4\sqrt{2}}{3}\right) \left(\frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, -\frac{2\sqrt{2}}{3}\right) \\ &= (1, -1, -2) - \left(\frac{2}{2}, -\frac{2}{2}, 0\right) - \left(\frac{4}{9}, \frac{4}{9}, -\frac{16}{9}\right) \\ &= (1-1, -1+1, -2-0) - \left(\frac{4}{9}, \frac{4}{9}, -\frac{16}{9}\right) \end{aligned}$$

$$\omega_3 = (0, 0, -2) - \left(\frac{4}{9}, \frac{4}{9}, -\frac{16}{9}\right) = \left(-\frac{4}{9}, -\frac{4}{9}, -\frac{2}{9}\right)$$

$$\|\omega_3\| = \sqrt{\frac{16}{81} + \frac{16}{81} + \frac{4}{81}} = \sqrt{\frac{36}{81}} = \frac{6}{9} = \frac{2}{3}$$

$$\therefore u_3 = \frac{\omega_3}{\|\omega_3\|} = \frac{\left(-\frac{4}{9}, -\frac{4}{9}, -\frac{2}{9}\right)}{\frac{2}{3}} = \frac{3}{2} \left(-\frac{4}{9}, -\frac{4}{9}, -\frac{2}{9}\right) = \left(-\frac{2}{3}, -\frac{2}{3}, -\frac{1}{3}\right)$$

$\therefore$ Orthonormal basis is $\{u_1, u_2, u_3\} = \left\{\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right), \left(\frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, -\frac{2\sqrt{2}}{3}\right), \left(-\frac{2}{3}, -\frac{2}{3}, -\frac{1}{3}\right)\right\}$

⑥ Let $\{e_1, e_2, \dots, e_n\}$ be an orthonormal basis of an inner product space V over $\mathbb{R}$. Show that for any $v \in V$

$$v = \langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2 + \dots + \langle v, e_n \rangle e_n$$

Sol Since $\{e_1, e_2, \dots, e_n\}$ is orthonormal basis so $\langle e_i, e_j \rangle = 0, i \neq j$
and $\|e_i\| = \langle e_i, e_i \rangle = 1, i = j$

Also since $\{e_1, e_2, \dots, e_n\}$ is basis for V , so any vector $v \in V$ can be written as linear combination of basis vectors.

Therefore $v = a_1 e_1 + a_2 e_2 + \dots + a_n e_n$ ——— ①

Taking inner product with $e_1, e_2, \dots, e_n$

$$\langle v, e_1 \rangle = \langle a_1 e_1 + a_2 e_2 + \dots + a_n e_n, e_1 \rangle$$

$$\begin{aligned} \langle v, e_1 \rangle &= a_1 \langle e_1, e_1 \rangle + a_2 \langle e_2, e_1 \rangle + \dots + a_n \langle e_n, e_1 \rangle \\ &= a_1 (1) + a_2 (0) + \dots + a_n (0) \end{aligned}$$

$$\boxed{\langle v, e_1 \rangle = a_1}$$

$$\begin{aligned} \langle v, e_2 \rangle &= a_1 \langle e_1, e_2 \rangle + a_2 \langle e_2, e_2 \rangle + \dots + a_n \langle e_n, e_2 \rangle \\ &= a_1 (0) + a_2 (1) + \dots + a_n (0) \end{aligned}$$

$$\boxed{\langle v, e_2 \rangle = a_2}$$

Similarly $\boxed{\langle v, e_3 \rangle = a_3}$

and so on $\boxed{\langle v, e_n \rangle = a_n}$

∴ ① becomes $v = \langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2 + \dots + \langle v, e_n \rangle e_n$

× ————— ×

Matrix of Linear Transformation:-

Let V & W be two finite dimensional vector spaces over the same field F . $\dim V = n$ & $\dim W = m$

Let $B = \{v_1, v_2, \dots, v_n\}$ and $E = \{w_1, w_2, \dots, w_m\}$ be any bases for V & W resp.

Any vector v in V can be expressed in a unique way as a linear combination of $v_1, v_2, \dots, v_n$ i.e. basis B .

$$\therefore v = a_1 v_1 + a_2 v_2 + \dots + a_n v_n$$

$$a_i \in F, i = 1, 2, \dots, n$$

a_i = coordinate vector of v relative to B

Let $T: V \rightarrow W$ be a linear transformation.

The images $T(v_1), T(v_2), \dots, T(v_n)$ are elements of W

Each image can be expressed uniquely as a linear combination of the basis vectors $w_1, w_2, \dots, w_m$.

$$T(v_1) = a_{11} w_1 + a_{21} w_2 + \dots + a_{i1} w_i + \dots + a_{m1} w_m$$

where $a_{ij} \in F$

$$T(v_2) = a_{12} w_1 + a_{22} w_2 + \dots + a_{i2} w_i + \dots + a_{m2} w_m$$

$$T(v_j) = a_{1j} w_1 + a_{2j} w_2 + \dots + a_{ij} w_i + \dots + a_{mj} w_m$$

$$T(v_n) = a_{1n} w_1 + a_{2n} w_2 + \dots + a_{in} w_i + \dots + a_{mn} w_m$$

The $m \times n$
Matrix of T is called
Matrix of linear
Transformation

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1j} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2j} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{i1} & a_{i2} & \dots & \boxed{a_{ij}} & \dots & a_{in} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mj} & \dots & a_{mn} \end{bmatrix}$$

a_{ij} is coefft of w_i in the image of $T(v_j)$

Q10(ii) If $T: V \rightarrow V$ is linear transformation show that $\langle T(e_j), e_i \rangle$ is the i th entry of the matrix representing T in the given basis $\{e_1, e_2, \dots, e_n\}$.

Sol $T: V \rightarrow V$ Let $\{v_1, v_2, \dots, v_n\} = \{w_1, w_2, \dots, w_n\} = \{e_1, e_2, \dots, e_n\}$

$$T(v_j) = a_{1j} w_1 + a_{2j} w_2 + \dots + a_{ij} w_i + \dots + a_{nj} w_n$$

$$T(e_j) = a_{1j} e_1 + a_{2j} e_2 + \dots + a_{ij} e_i + \dots + a_{nj} e_n$$

$$\begin{aligned} \langle T(e_j), e_i \rangle &= a_{1j} \langle e_1, e_i \rangle + a_{2j} \langle e_2, e_i \rangle + \dots + a_{ij} \langle e_i, e_i \rangle + \dots + a_{nj} \langle e_n, e_i \rangle \\ &= 0 + 0 + \dots + a_{ij} (1) + 0 + \dots + 0 \end{aligned}$$

$\because \{e_1, e_2, \dots, e_n\}$ is orthonormal basis

$$\langle T(e_j), e_i \rangle = a_{ij} = i\text{th entry in the matrix of } T$$

7.1-12

20) Let W be a subspace of an inner product space V . Show that there is an orthonormal basis of W which is part of an orthonormal basis of V .



Sol W is subspace of inner product space $V(R)$

Let $\{w_1, w_2, \dots, w_t\}$ be a basis for W

$$\dim(W) = t$$

$$\text{Let } \dim(V) = n$$

As $\{w_1, w_2, \dots, w_t\}$ is a linearly independent set of vectors in V (basis for W)

So by theorem "Any linearly independent set of vectors in a finite dimensional vector space V can be extended to a basis for V "

Another basis for $V = \{w_1, w_2, \dots, w_t, u_1, u_2, \dots, u_{n-t}\}$

$\Rightarrow$ Basis for $W \subseteq$ Basis for V — (1)

Transforming by Gram Schmidt process into orthonormal basis for W & V

(1) becomes orthonormal basis for $W \subseteq$ orthonormal basis for V .

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